

REFLOWECONOMY · DESIGN DECISIONS REGISTER

ReflowEconomy design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Approved follow-ups carried into the design: feed hood, self-closing hinges, tray padding; Value engineering restated at USD 25,505
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for all six open decisions (2026-10-02); RFE-DDR-003 accepted; moved to decisions made
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open decisions from RFE-DDR-001 to RFE-DDR-003; equipment cost treated as a value-engineering target

ReflowEconomy design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The structural engineer's seam beam section, fixings and propping sequence fit the 165 x 305 mm space reserved under the roof	The tray crossing and the aisle headroom are set from it	RFE-DDR-003, P1
2	The shredder's size, hopper position and top height (about 1,205 x 550 x 1,512)	The enclosure clears it by only 28 mm above, and the chute sits over the hopper	RFE-DDR-003, P5
3	The gearmotor stays within its rated temperature with the enclosure closed through a shift	The lined enclosure has no ventilation; add a lined vent baffle if it runs hot	RFE-DDR-003, P5
4	The extruder frame top is flat at about 900 mm with room for six bolts, and the barrel axis is where hood B's hole is	Hood B is made to fit the frame and barrel	RFE-DDR-003, P2
5	Both press platens are at the same working height (taken as 950 mm) and a 1 m mould fits the cooling press and the 1.04 m second opening	The transfer bridge and the booth openings are set from them	RFE-DDR-003, P4
6	The fan and filter box size, mass, inlet height (2.05 m) and outlet position	The fan stand and the stack are made to fit it	RFE-DDR-003, P8
7	The board's size fits its 700 x 800 mm place on the backboard	The backboard is cut before the board is chosen	RFE-DDR-003, P6

■ Value engineering

Value-engineering target: USD 25,000. Estimated cost of the constructable design: USD 25,505 (USD 505 over the target).

The target covers equipment items 1 to 12 (requirement R10; a hypothetical control target, not a limit; `budget_usd` is null because this is a playbook repo). The container shell adds about USD 9,200; it is site-dependent and outside the target. Main cost drivers and savings worth trying:

- The largest lines are the sheet press, cooling press and transfer bridge (USD 6,250), the shredder with its enclosure and feed hood (USD 4,205), the extruder (USD 4,000) and the fume extraction (USD 3,350). Together they are 70 % of the equipment.

- Making the design constructable added USD 1,100 to the equipment (sash, take-off, brackets, wall sleeve and fan stand USD 400; backboard and cable tray USD 400; transfer bridge USD 250; drying fan stand USD 50) and USD 1,200 to the container shell (seam beam, floor plate, flashing).
- Carrying the decisions of 2026-10-02 into the design added USD 205: the cranked feed hood and three self-closing door hinges (USD 105), two pairs of self-closing cage gate hinges (USD 60), and padding and hazard tape on the tray crossing (USD 40).
- Savings worth trying: a cooling press built from scrap steel plate and the drying rack and fan stand from offcuts of the same stock (about USD 300 together); second-hand pallet racking (about USD 150); a rented shed instead of containers, which removes the USD 1,200 of seam work and most of the shell cost (a shell saving, outside the equipment target, so it does not reduce the USD 505 gap); quotes from two local fabricators for the press, which is the largest single line.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	Plastics-only reference line with aluminium as a separate add-on bay; two 40 ft container footprints or a shed of about 60 m ² ; PET sold as clean flake; passport schema v0.2 with example records; single-phase supply with staggered heating; keep CERN-OHL-S-2.0 on the playbook text for now	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	RFE-DDR-001, items 1 to 6
2026-09-25	First release licensing (documents CC BY-SA 4.0, hardware CERN-OHL-S-2.0, scripts MIT); R8 keeps the honest total with an intake quality rule (R16, 10 % residue threshold); derated 5 kW single-phase sheet press as the reference; 1.5 kW fume fan; rented shed preferred, container side walls removed only to a structural engineer's design	Amish: "i accept all your recommendations, go with them across all repos."	RFE-DDR-002, items 6 (release) and 8 to 11
2026-09-25	TRL 4 on hold for every repo	Amish: "Make sure we don't proceed to TRL 4 on any of them."	RFE-DDR-001
2026-10-01	Budgets are value-engineering targets, not limits; R10 reported as over or under its target	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens"	This register; RFE-CAL-001 v0.3
2026-10-02	Design for construction accepted: all the changes of Table 1 and their knock-on changes, as made, with the four safety-case items decided separately (A1 to A4, below)	Amish: "i approve your recommendations for all 555 open decisions."	RFE-DDR-003, Table 1
2026-10-02	Shredder feed: keep the 1.70 m chute mouth, but do the ISO 13857 reach check from the model now rather than at the safety review, and lengthen or baffle the chute until it passes	Amish: "i approve your recommendations for all 555 open decisions."	RFE-DDR-003, A1
2026-10-02	Press booth: one sliding sash over the two openings, as modelled	Amish: "i approve your recommendations for all 555 open decisions."	RFE-DDR-003, A2
2026-10-02	Shredder enclosure door and cage gates: hinged, opening into the aisle only during locked-off maintenance or loading, with self-closing hinges and the rule in the playbook's safety section; sliding door and gates instead if the open door leaves less than the local code's minimum escape width	Amish: "i approve your recommendations for all 555 open decisions."	RFE-DDR-003, A3
2026-10-02	Aisle headroom: accept 2.04 m, confirmed against the local code by the safety professional, with the tray crossing padded and marked with hazard tape	Amish: "i approve your recommendations for all 555 open decisions."	RFE-DDR-003, A4
2026-10-02	First co-design partner and region: (a), a waste-picker cooperative; first candidate to approach, SWaCh in Pune, India, reached through the WIEGO network (not agreed)	Amish: "i approve your recommendations for all 555 open decisions."	RFE-DDR-001, item 7; RFE-DDR-002